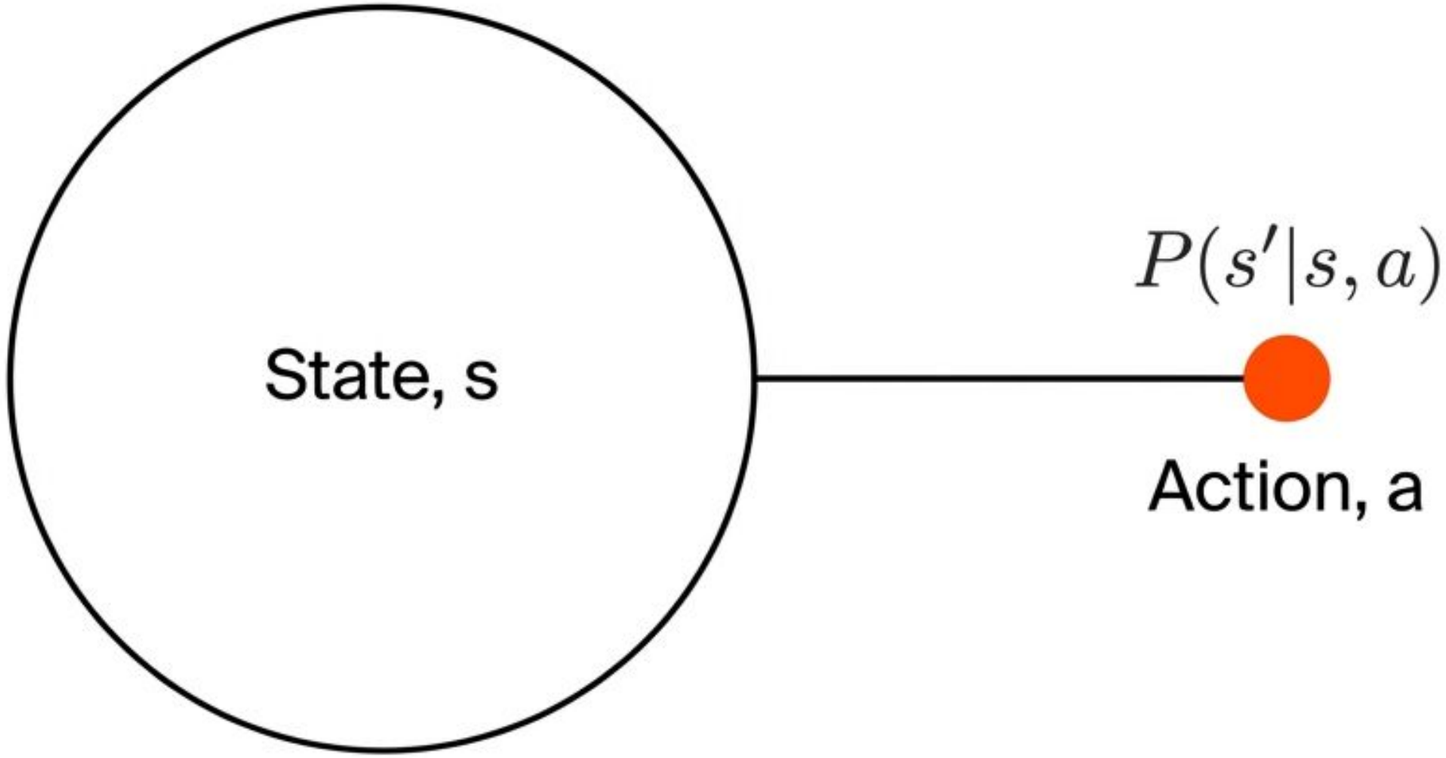
Finite Markov Decision Processes

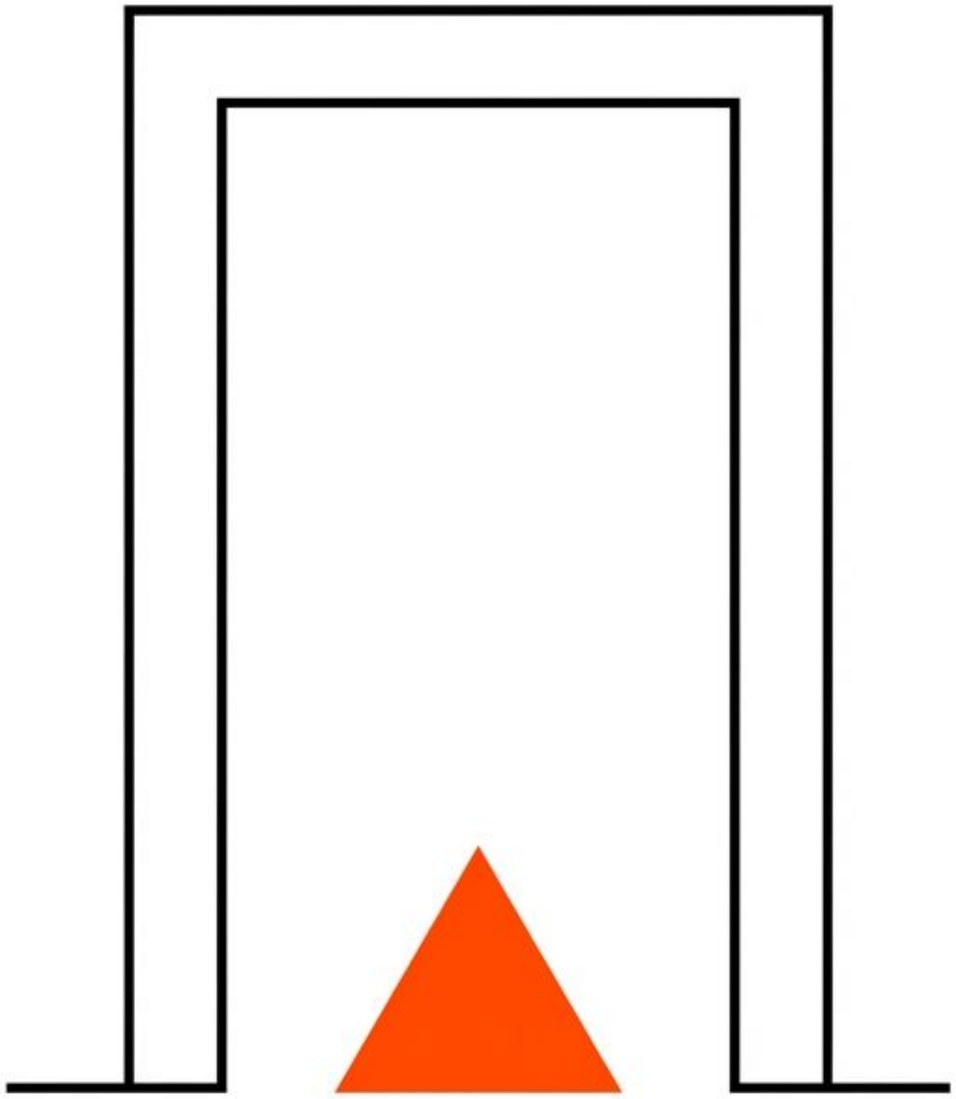
The Agent, The Goal, and The Math of Long-Term Value



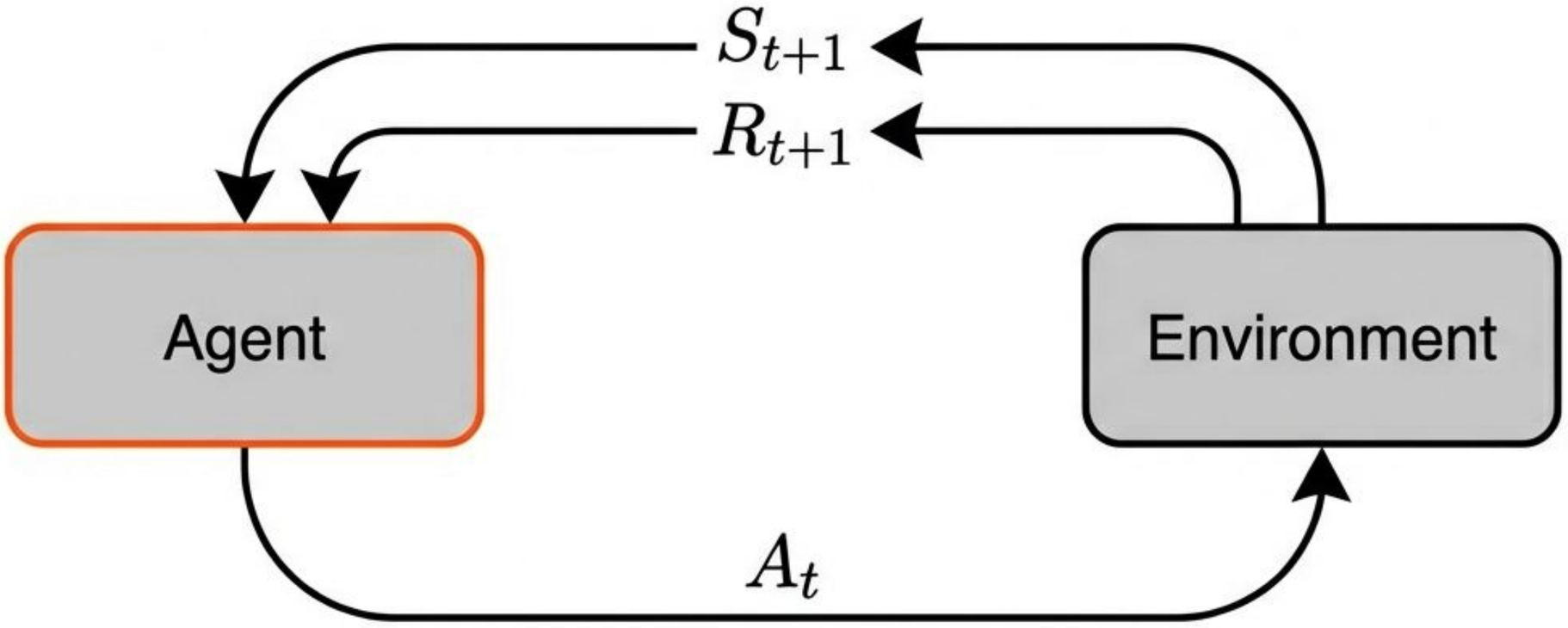
Learning from Interaction

Reinforcement learning is not about being told what to do. It is about discovering which actions yield the most reward by trying them.

- **Trial-and-Error Search**
- **Delayed Reward**

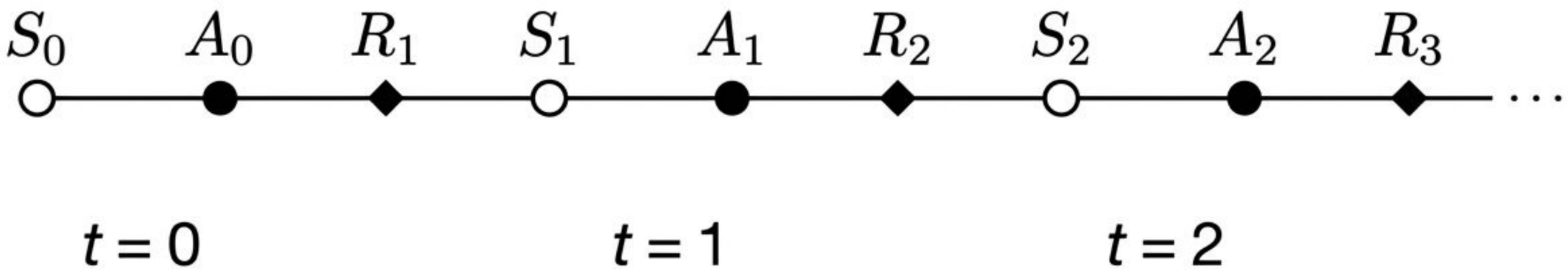


The Agent-Environment Interface



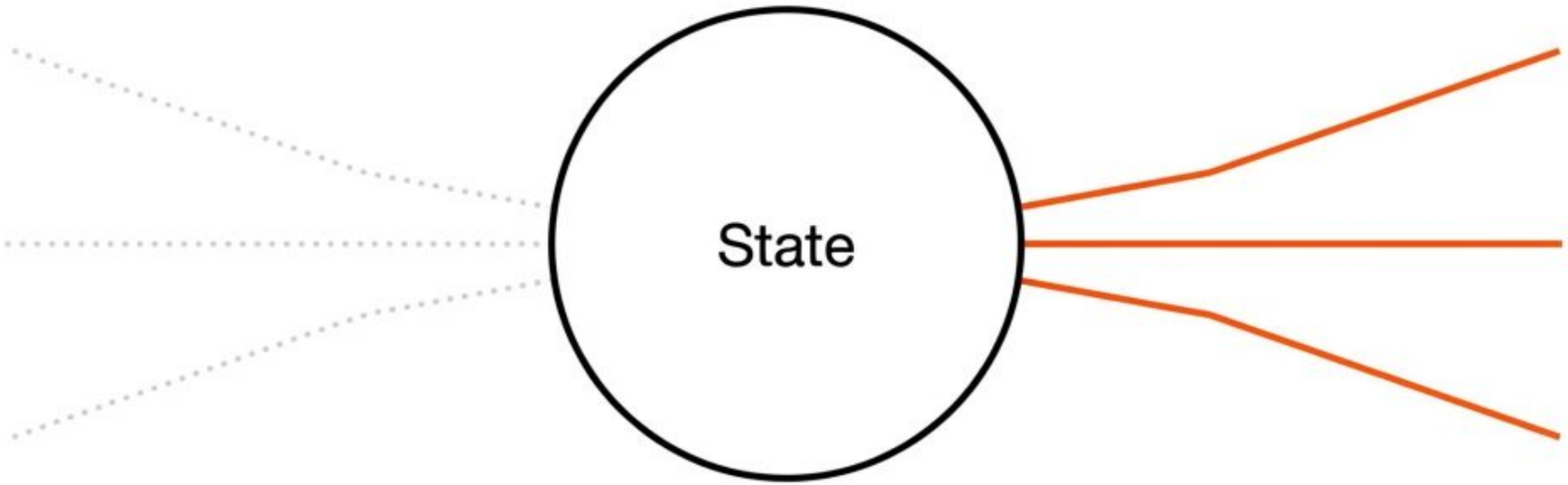
The learner (Agent) interacts with the world (Environment) at discrete time steps t.

The Trajectory



$$S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3 \ldots$$

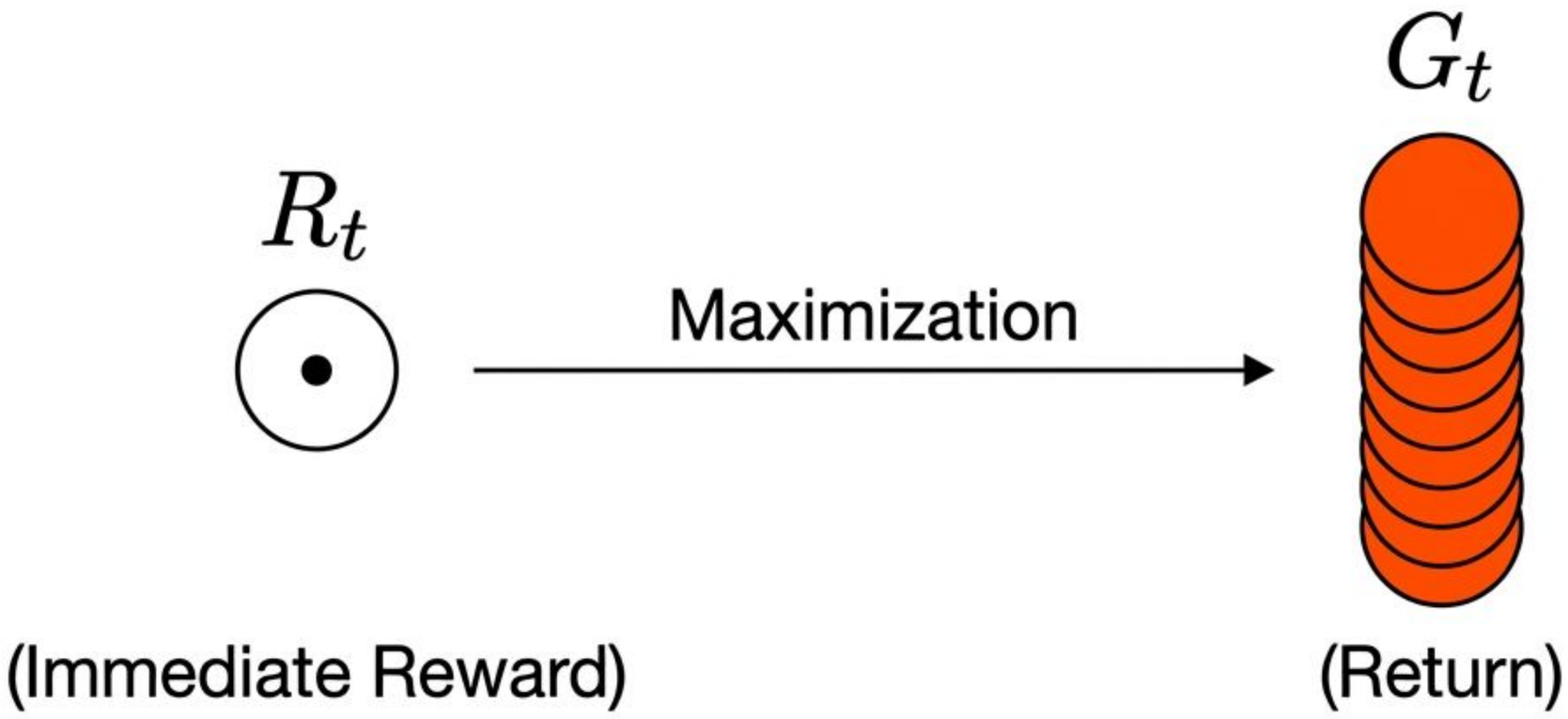
The Markov Property



The state summarizes the past compactly.

$$p(s', r|s, a)$$

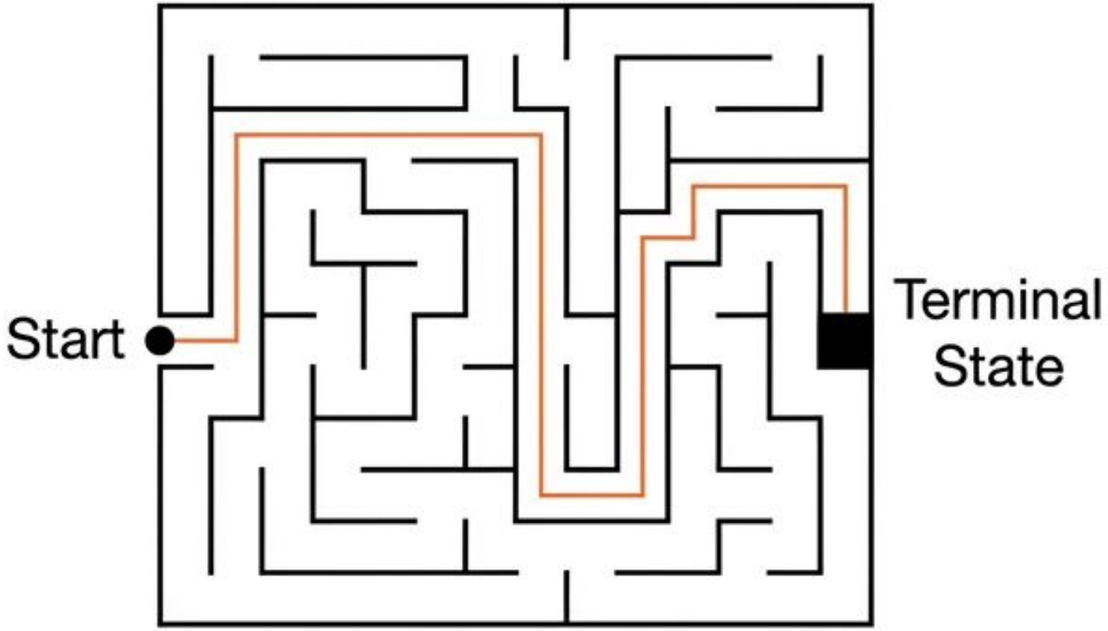
The Reward Hypothesis



All goals can be framed as the maximization of the expected value of the cumulative sum of a received scalar signal (reward).

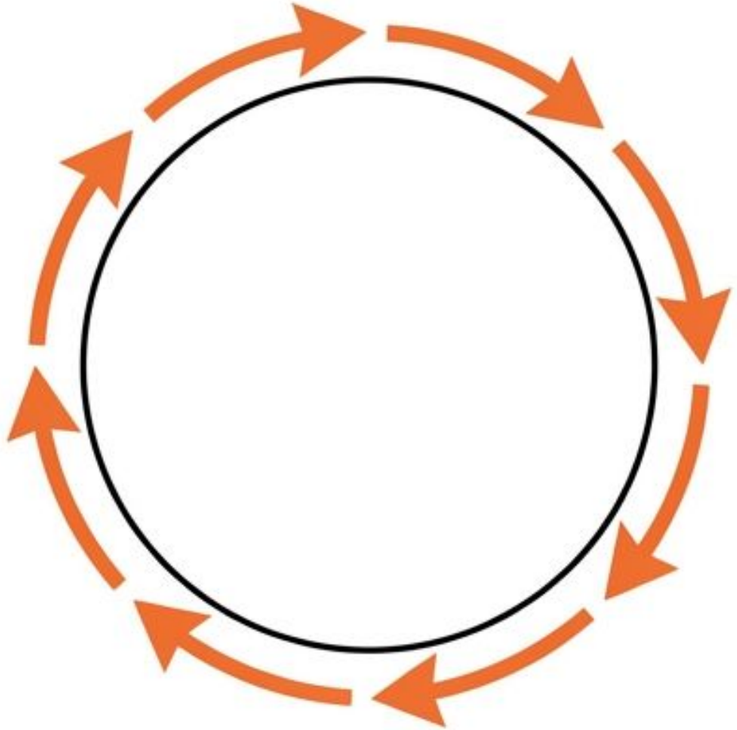
Episodic vs. Continuing Tasks

Episodic



Interaction breaks into subsequences.

Continuing



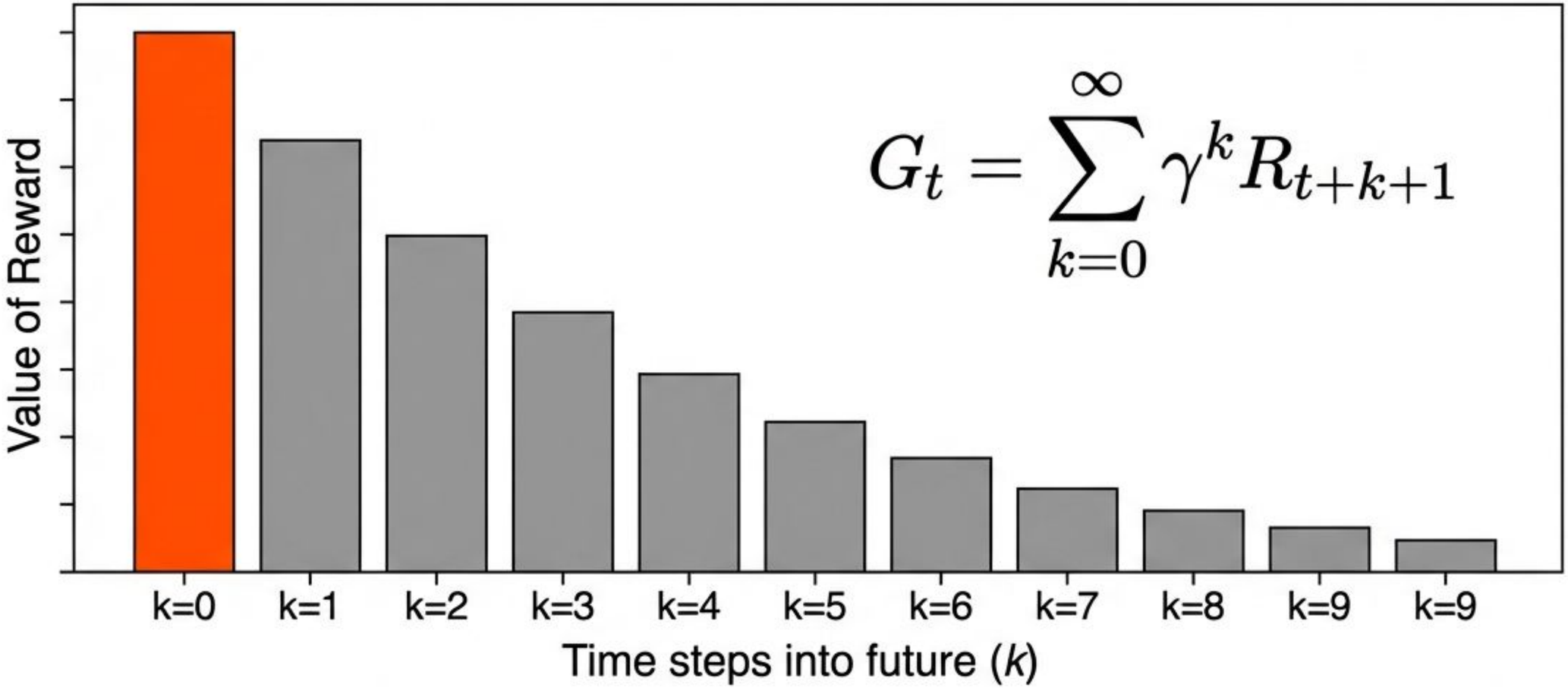
Interaction goes on continually ($T = \infty$).

The Return (G_t)

$$G_t = \overbrace{R_{t+1} + R_{t+2} + R_{t+3} + \ldots + R_T}^{\text{The Goal}}$$

The Return is the total accumulated reward from time t.

Discounting the Future (γ)



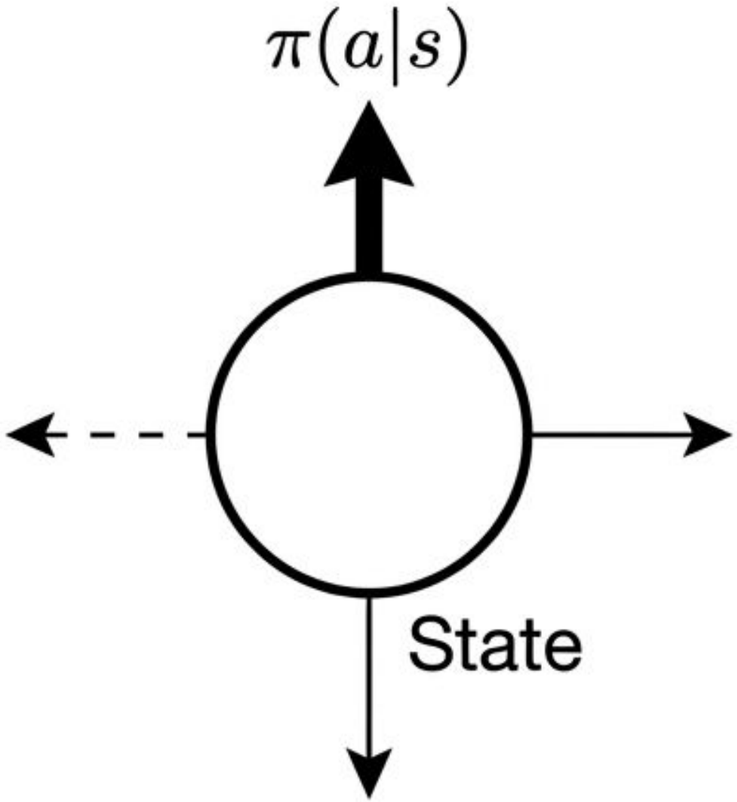
$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

$\gamma = 0$ is myopic. $\gamma \to 1$ is farsighted.

From Rewards to Value

A state can yield low reward but have high value (or vice versa) based on what follows it.

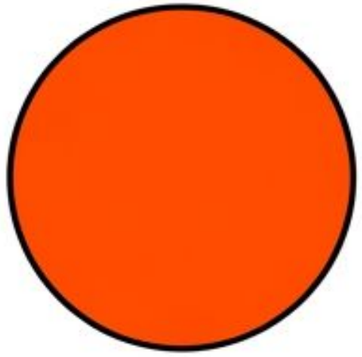
The Policy (π)



A policy is a mapping from states to probabilities of selecting each possible action.

State-Value Function (v_π)

$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$$

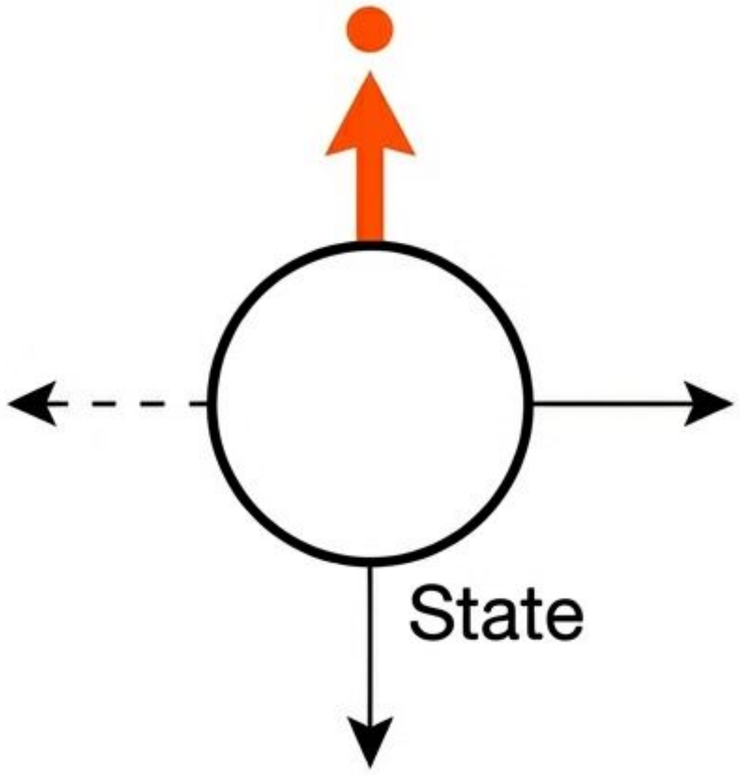


State s

How good is it to be in state s?
The expected return starting in s and following policy π.

Action-Value Function (q_π)

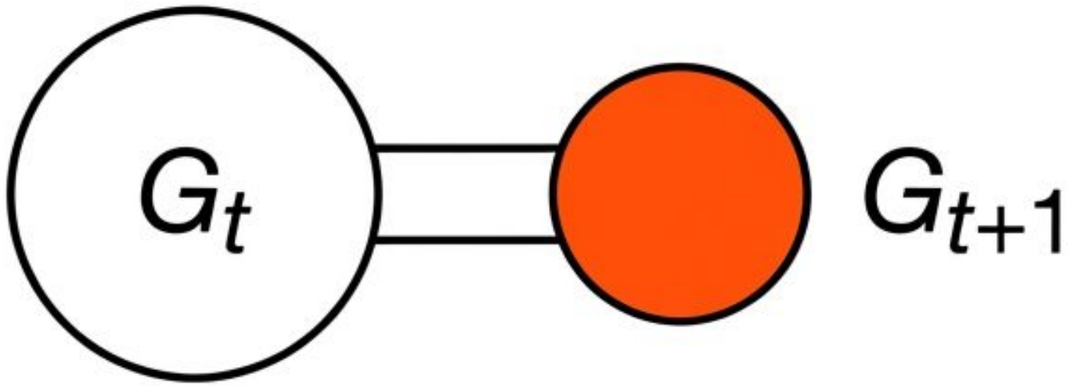
$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$$



How good is it to take action a in state s?

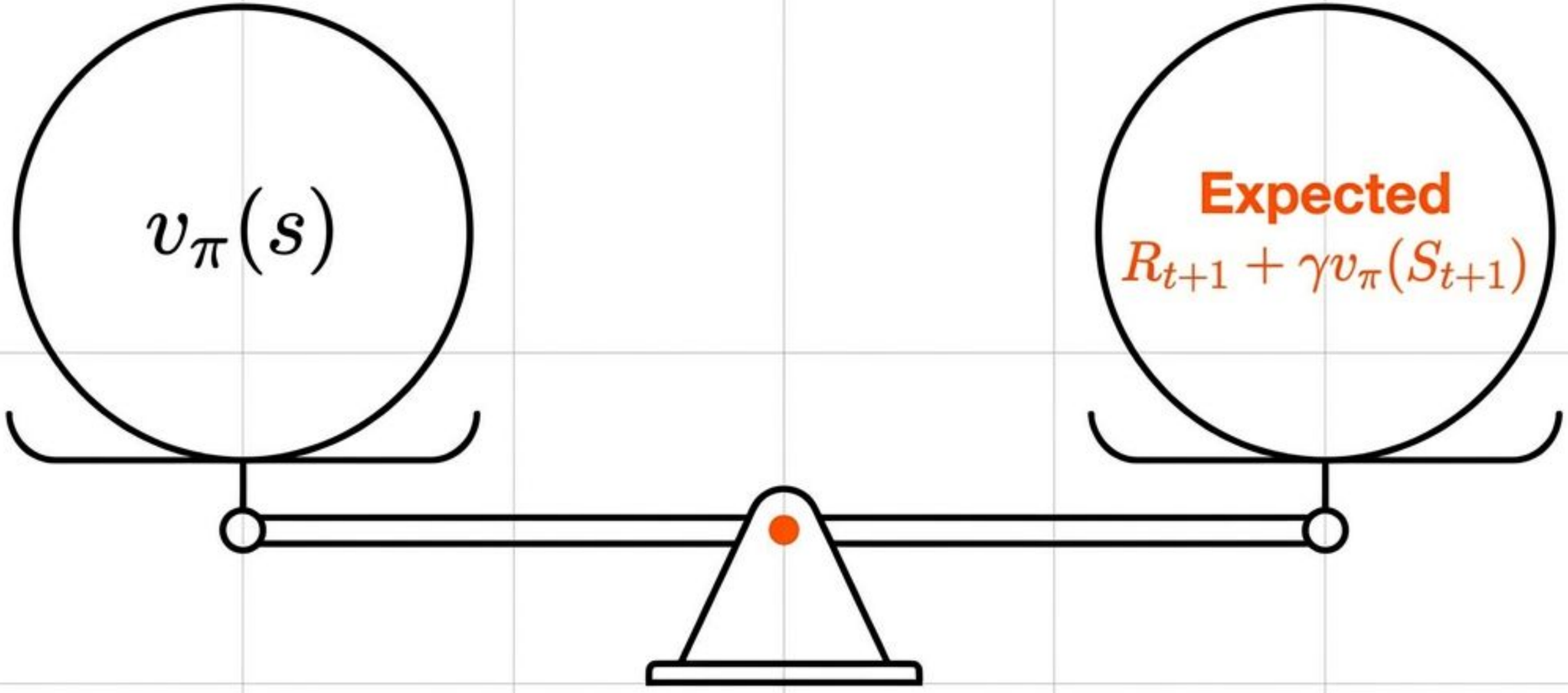
The Recursive Nature of Returns.

$$G_t = R_{t+1} + \gamma G_{t+1}$$



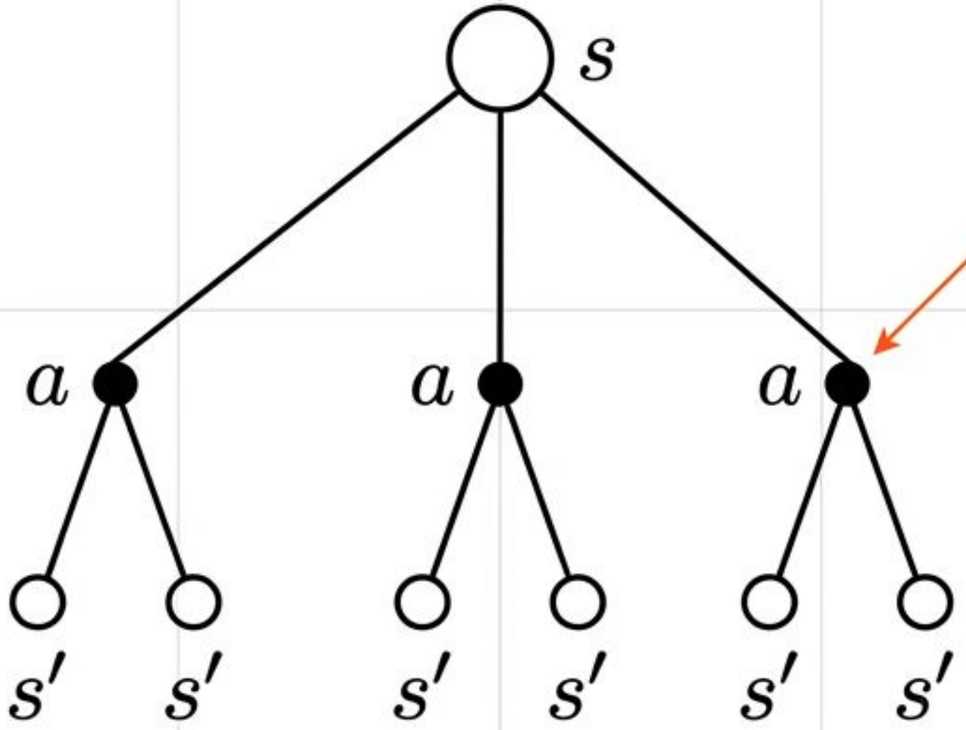
The value of ‘now’ is the immediate reward plus the discounted value of ‘later’.

The Bellman Equation for v_π



A consistency condition: Value of Start = Expected Value of Next Step

Visualizing the Look-Ahead



Averaging over actions (π) and environmental dynamics (p).

The Backup Diagram

The Full Bellman Equation

$$v_\pi(s) = \underbrace{\sum_a \pi(a|s)}_{\text{Average over Policy}} \underbrace{\sum_{s',r} p(s', r \mid s, a)}_{\text{Average over Dynamics}} \underbrace{[r + \gamma v_\pi(s')]}_{\text{Recursive Target}}$$

Optimal Policies

A policy π is better than π' if its expected return is greater for all states.

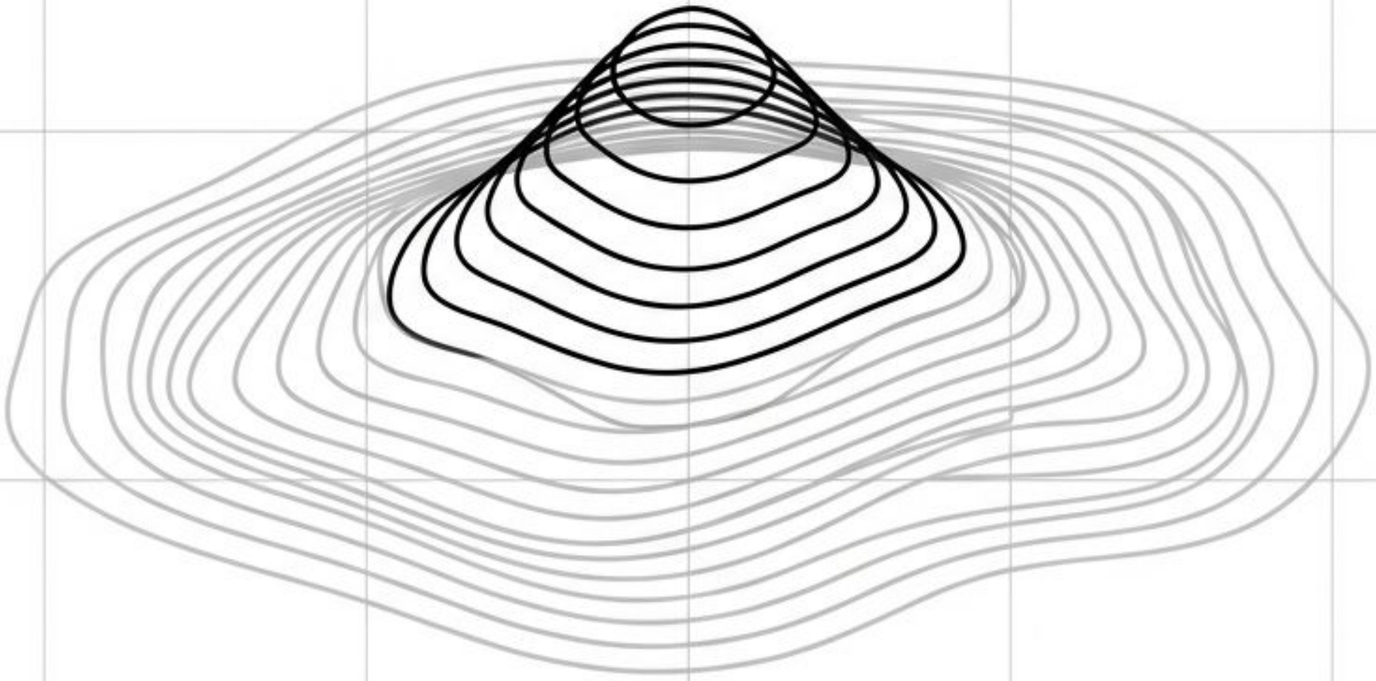
$$\pi \geq \pi' \text{ if } v_\pi(s) \geq v_{\pi'}(s) \text{ for all } s.$$

There is always at least one policy that is better than or equal to all others: The Optimal Policy (π_*).

The Optimal Value Functions

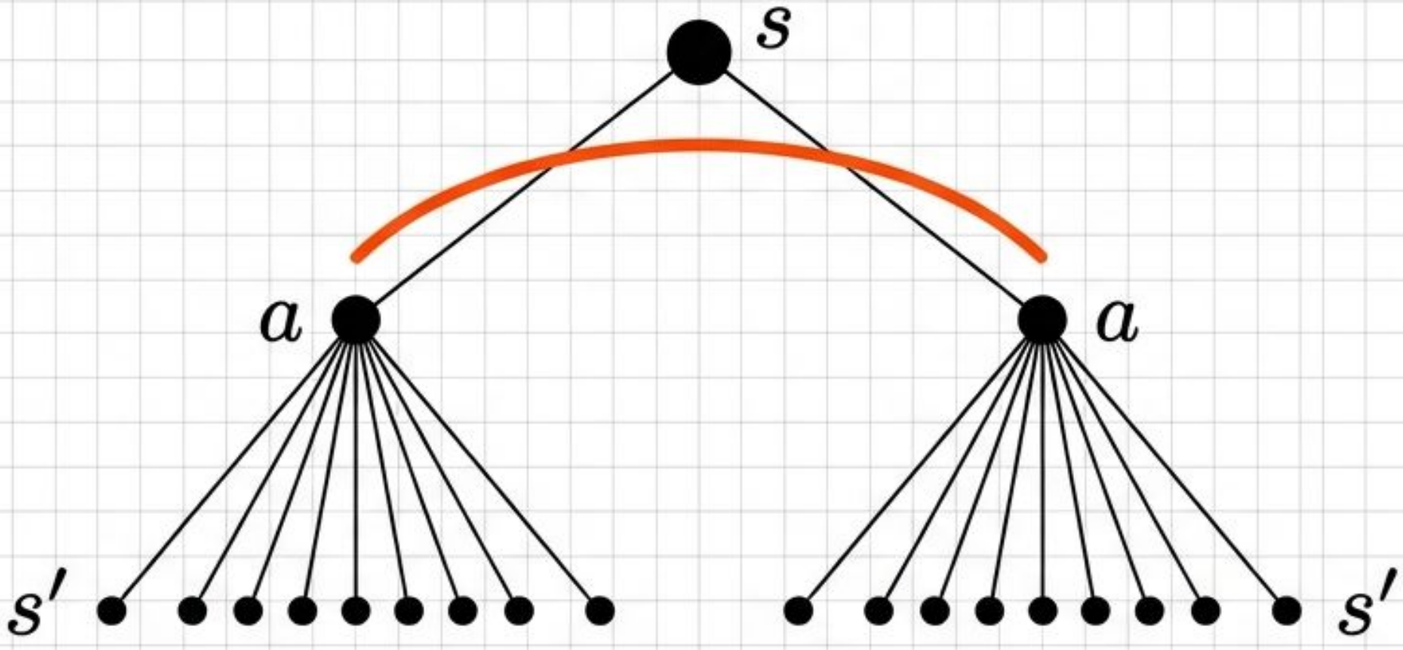
$$v_*(s) = \max_{\pi} v_\pi(s)$$

$$q_*(s, a) = \max_{\pi} q_\pi(s, a)$$



The Bellman Optimality Equation

$$v_*(s) = \max_a \mathbb{E}\left[R_{t+1} + \gamma v_*(S_{t+1})\right]$$



Greed is Optimal.

14.4 →	16.0 →	17.8 →	19.1 →	20.2
15.2 →	18.0 →	19.5 →	21.3 →	22.4
16.9 →	19.2 →	21.6 →	22.9 →	23.8
17.5 →	20.4 →	22.7 →	23.1	24.2 ↓
19.0 →	21.1 →	23.5 →	24.0 →	24.5

If you have v_*, the optimal policy is simple: be greedy.